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Enhancing Heterogeneous Graph Learning with Semantic-Aware Meta-Path Diffusion and Dual Optimization

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ABSTRACT

Heterogeneous graph learning aims to extract semantic and structural information from multiple node types, edges, and meta-paths, learning low-dimensional embeddings that preserve core characteristics to support downstream tasks. To address the core challenges of insufficient semantic mining and weak learning synergy in heterogeneous graph learning, this paper proposes a heterogeneous graph learning method integrating Semantic-aware Meta-path perturbation and Collaborative Dual-learning optimization(SMCD). First, the method constructs auxiliary meta-paths based on the original meta-paths, and then designs two augmentation schemes to generate augmented views: For semantic-level augmentation, it performs edge perturbation based on semantic similarity, and enhances the semantics of core meta-paths with the semantics of auxiliary meta-paths via a diffusion model; For task-level augmentation, it utilizes a diffusion model and semantic weights to select the top-k semantically relevant nodes for each node in the core meta-path graph, reconstructing the meta-paths graph structure. Then, a two-stage attention aggregation graph encoder is adopted to output the final node embeddings. Finally, a self-supervised and supervised (i.e., Dual-learning) collaborative optimization strategy that flexibly adapts to label distribution is used to optimize the objective-this not only balances the discriminability and generality of representations but also adapts to scenarios with different degrees of label scarcity. Experimental results on three public datasets illustrate that our proposed method achieves remarkable advantages in both node classification and node clustering tasks. Our datasets and source code are available.[1]

1. Introduction

Heterogeneous graph learning is a learning paradigm for heterogeneous graphs containing multiple types of nodes and edges. Learns node/graph representations by fusing structural and semantic information to support downstream tasks such as node classification [1–3], link prediction [4], and recommendation systems [5,6]. In recent years, numerous studies [7–12] have shown that graph neural networks (GNNs) can better learn heterogeneous graph representations. Heterogeneous graph neural networks (HGNNs) [13] are models specifically designed to handle heterogeneous graphs. Through type-adaptive information aggregation and semantic fusion, they map nodes/graphs into low-dimensional representations to support downstream tasks. However, most mainstream HGNN models follow a supervised learning [14,15] paradigm, and their effective training is highly dependent on large-scale and high-quality labeled data. The acquisition of such data often requires significant manual annotation costs. In practical scenarios such as academic networks, industrial recommendation systems, and knowledge graphs, it is either difficult to fully annotate due to the large data scale or extremely challenging to obtain labels due to the high professionalism of tasks, which severely limits the practical application of HGNNs.

To alleviate the burden of label annotation, researchers have introduced self-supervised [16–19] learning into the field of heterogeneous graphs. This paradigm automatically mines supervisory signals from unlabeled data, enabling models to learn effective representations even in scenarios with few or no labels. However, most of these methods focus on feature extraction from unlabeled nodes and fail to fully utilize the supervisory information contained in the small number of labeled nodes that exist. As a result, the models' representations lack a clear task orientation. To address this issue, the CALHG [20] model proposes a fusion framework that combines self-supervised auxiliary learning with supervised learning, leveraging both the guiding role of labeled nodes and the supplementary value of unlabeled nodes. Nevertheless, its design-with supervised learning as the main component and self-supervised learning as the auxiliary-has obvious limitations in scenarios where labeled data is extremely scarce. On one hand, the limited labeled data

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[1] https://github.com/Ding-guang-hua/SMCD

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cannot provide sufficient supervisory signals to guide the model in learning task-relevant discriminative representations, making it difficult for supervised learning (as the dominant component) to play an effective role. On the other hand, the auxiliary self-supervised learning is not assigned sufficient optimization weight, which restricts its potential to mine structural and semantic invariance information from unlabeled data and fails to effectively compensate for the representation learning defects caused by insufficient labels. Therefore, how to design a reasonable fusion mechanism to achieve the collaborative optimization of supervised learning and self-supervised learning (Abbreviated as "Dual-learning")-ensuring that the learned representations are both discriminative and general, while adapting to scenarios with different degrees of label scarcity-has become a key urgent problem to be solved in the current field of heterogeneous graph learning.

To effectively learn node representations in heterogeneous graphs, existing methods [17,21,22] rely mainly on meta-paths to expand raw data for mining semantic information. As a relational sequence connecting two types of nodes, meta-paths [23] serve as the core carrier for the semantic representation of heterogeneous graphs. Different meta-paths correspond to distinct semantic correlations. The denser the meta-path instances between nodes, the stronger their inherent relevance and semantic consistency [24]. Therefore, graph augmentation should prioritize preserving such high-correlation connections, avoiding undifferentiated edge modifications that may destroy critical semantic structures and cause semantic distortion. To refine the granularity of semantic mining, this paper explicitly defines core and auxiliary meta-paths: core meta-paths are basic semantic carriers selected based on task priors or domain expertise, while auxiliary meta-paths are relation sequences in the heterogeneous graph that are not predefined as core paths but contain complementary semantics. These auxiliary meta-paths can enrich the semantic expression of core meta-paths from multiple dimensions and enhance the comprehensiveness of representations. However, existing meta-path-based graph augmentation schemes have obvious limitations: On one hand, they only perform simple edge perturbation on meta-path-induced graphs, without considering the semantic similarity between nodes. This easily leads to the accidental removal of connections between nodes with high semantic relevance, damaging key semantic structures. On the other hand, these schemes over-rely on the inherent semantics of pre-defined meta-paths (i.e., the core meta-path) and fail to fully exploit additional semantic information carried by other auxiliary meta-paths, which limits the richness of representation learning. Therefore, designing a semantic-aware meta-path perturbation scheme that balances semantic similarity and low computational cost while enhancing the semantic expression of core meta-paths by fusing the semantics of auxiliary meta-paths remains a key urgent problem to be solved in the current field of heterogeneous graph learning.

To address the above two challenges, this paper proposes a heterogeneous graph learning method that integrates Semantic-aware Meta-path perturbation and Collaborative Dual-learning optimization(SMCD). As illustrated in Fig. 1. First, based on the original meta-paths, a set of auxiliary meta-paths is devised to enhance the core meta-paths. Then, leveraging the core meta-paths and auxiliary meta-paths, this paper proposes two augmentation strategies for constructing augmented views. The first is a semantics-level augmentation, which performs edge perturbation based on semantic similarity across all meta-path graphs, and utilizes the semantics of auxiliary meta-paths to enhance the semantics of core meta-paths via a diffusion model. The second scheme is task-level augmentation, which utilizes the diffusion model and semantic weights to select the top-k nodes with the strongest semantic relevance for each node in the core meta-path graph, constructs local neighborhoods, and adjusts the original graph structure. Subsequently, the model adopts a graph attention layer for each augmented meta-path graph to obtain node embeddings. It then automatically learns the significance of different meta-path graphs via graph-level attention and fuses the corresponding node embeddings to generate the final embeddings. Finally, a self-supervised and supervised collaborative optimization strategy is proposed. This strategy flexibly adjusts the weights of self-supervision and supervision according to the label distribution characteristics of the dataset, guaranteeing that the learned representations are both discriminative and general, while adapting to scenarios with different degrees of label scarcity.

To conclude, the key contributions of this work are summarized below:

- We proposed SMCD, a novel heterogeneous graph learning method, integrating both semantic-aware meta-path perturbation and self-supervised–supervised collaborative optimization. Building on the original graph, we proposed two graph data augmentation strategies. First, a collection of meta-paths is designed as auxiliary meta-paths, edge perturbation is performed on meta-path graphs based on semantic similarity, and the semantic information of auxiliary meta-paths is employed to strengthen the semantics of core meta-paths. Second, the core meta-path graph is reconstructed with the aid of a diffusion model.
- Additionally, we proposed a self-supervised and supervised collaborative optimization strategy for SMCD. This strategy flexibly adjusts the weights of self-supervision and supervision based on the label distribution traits of the dataset to boost the performance of supervised or self-supervised HGNNs.
- Comprehensive experiments and analyses of SMCD-HGL are carried out on three public datasets, demonstrating its significant advantages in node classification and node clustering tasks.

The outline of this paper is as follows. We provide an overview of existing related works in Section 2 and formally define the research problem in Section 3. Section 4 provides a detailed introduction to our model. The experimental results are presented and discussed in Section 5.To conclude, we discuss the findings, draw conclusions, and propose possible future research in Section 6.

2. Related Work

Heterogeneous Graph Neural Network. During the evolution of Heterogeneous Graph Neural Networks (HGNNs), we simply categorize the models into three types based on their approaches to utilizing heterogeneous information and core technical paths. The first category is attention-driven fine-grained modeling. For example, HAN [25] distinguishes node relevance and meta-path importance through two attention layers (node-level and semantic-level); HetSANN [26] aggregates multi-relational information of projected neighbors via attention; HGT [27] characterizes heterogeneous attention using node and edge type-dependent parameters. All of these models improve the accuracy of representations by dynamic weight assignment. The second category is meta-path and multi-view fusion. This includes MAGNN [9], which integrates intermediate nodes within meta-paths and multi-path information through three core components: node content conversion, intra-meta-path aggregation, and inter-meta-path integration; HAGNN [10], which coordinates meta-path neighbors and direct neighbors via a two-stage aggregation ("intra-type + inter-type"); and GTN [28], which automatically generates meta-paths to construct new graph structures. All these schemes focus on mining heterogeneous graph semantics around meta-paths. The third category is prior-free and multi-information fusion. RSHN [29] merges the original graph and the coarsened line graph to achieve mutually enhanced node-edge embeddings; HetGNN [11] samples strongly correlated heterogeneous neighbors and performs grouped aggregation; MHGCN [30] learns the interaction of meta-paths with different lengths through multi-layer convolution. All of these models aim to reduce the reliance on manual priors. To address the core challenges in heterogeneous graph learning, several representative models have proposed innovative solutions: BPHGNN [31] introduces Basic Behavioral Patterns (BBP), which accurately characterizes the complex interactions in multi-relational heterogeneous graphs by enumerating non-empty relation combinations, thereby alleviating the

over-smoothing issue; DAG-HFC [32] focuses on feature scarcity and semantic ambiguity, achieving feature completion and semantic enhancement through topological optimization and a dual-domain attention mechanism; AdaGCLa [33] targets the topological bias in large-scale training, dynamically adjusting perturbation parameters via adaptive graph augmentation and a Node-centric Information Bottleneck (Node-IB), which balances similarity and diversity during batch sampling training while ensuring both training efficiency and generalization. Additionally, although PGR-MOOD [34] is primarily designed for out-of-distribution (OOD) detection in molecular graphs, its idea of prototype graph generation driven by diffusion models provides valuable insights for semantic consistency modeling in heterogeneous graphs. In terms of input quality and scale optimization, Hu et al. [7] used priority sampling to select high-quality path instances; HGT [27] is paired with HGSampling to adapt to ultra-large-scale graphs. Although these models have made respective breakthroughs in heterogeneous graph representation learning, some schemes still face issues such as insufficient depth in semantic information mining, difficulty in adapting perturbation strategies to meta-path characteristics, and weak synergy between structure and semantics during the learning process.

Graph Contrastive Learning. Focusing on the contrastive learning paradigm, existing graph representation learning models can be categorized into two types. The first type is for homogeneous graphs, which mine structural and attribute information through view construction and contrastive objective optimization. For example, DGI [35] utilizes graph convolutional networks to generate patch representations and high-order graph summaries, with its core approach accomplishing unsupervised learning through maximizing mutual information between the two. GRACE [36] adopts a hybrid scheme at the structural and attribute levels to generate dual views, learning embeddings through maximizing the coherence of node representations. GCA [37] incorporates topological and semantic priors to design an adaptive augmentation scheme: at the structural level, it highlights important connections based on node centrality; at the attribute level, it enhances semantic recognition through differential noise addition. The second type is for heterogeneous graphs, which focus on addressing semantic diversity and structural complexity. Representative works include STENCIL [38], which generates multi-semantic views based on meta-paths, extracts information with a novel multi-view contrastive aggregation objective, and introduces structural embeddings to mine real hard negative samples. CPT-HG [39] designs relation-level and subgraph-level semantic-aware pre-training tasks, combining contrastive learning to capture heterogeneous graph semantics and structural properties while enabling knowledge transfer. HeCo [40] constructs dual views of network patterns and meta-paths, optimizing embedding performance through cross-view contrast and high-quality hard negative sample generation. Both HGCML [41] and HGCMA [17] generate multi-views based on meta-paths: HGCML [41] improves the accuracy of contrastive learning through positive sampling strategy optimization; HGCMA [17] enhances the robustness of contrastive learning via topological and semantic fusion, and further strengthens the robustness of node representations through multi-view construction and cross-view contrast. DMGI [42] fuses multi-relational embeddings with consistency regularization and a general discriminator, while inferring relation importance through an attention mechanism. CKD [43] focuses on meta-path relevance, learning better embeddings through collaborative knowledge distillation at the regional and global granularities. In addition, CALHG [20] encodes rich information of heterogeneous graphs through contrastive auxiliary learning to improve downstream task performance.

Although existing contrastive learning models have made respective breakthroughs, they still suffer from inadequacies in the depth of heterogeneous graph semantic mining, the accurate utilization of meta-path semantics, and learning synergy. To address these limitations, this paper adopts semantic-aware meta-path perturbation and dual-learning collaborative optimization, which adapts to the characteristics of heterogeneous graphs, enhances the robustness and generalization of representations, and breaks through the bottlenecks of traditional methods.

Diffusion Model. Diffusion models, leveraging their core mechanism of injecting reasonable noise through random perturbation and recovering core semantics via denoising reconstruction, have provided novel solutions to key challenges in graph learning-such as semantic propagation modeling, structural robustness enhancement, and discriminative representation learning. Existing works have conducted in-depth explorations around the integration of diffusion models and graph representation learning, forming diverse technical pathways. GPS [44] focuses on the connection between self-supervised graph learning and graph diffusion. It selects high-quality positive samples consistent with the semantic propagation characteristics of diffusion models through four complementary metrics-node centrality, topological distance, neighborhood overlap, and semantic distance-and alleviates sampling bias with fusion-based contrastive objectives. GFT [45] expands the discriminability of tokens through the orthogonal regularization of computation tree vocabulary, reconstructs computation trees by virtue of denoising mechanisms, and learns structure-invariant features via generative objectives. On the other hand, GIT [46] captures the corruption-invariant semantics of task-trees through dual-view reconstruction tasks, where diffusion models provide technical support for maintaining the structural integrity of task-trees, thereby enhancing the robustness of representations. Furthermore, G2PM [47] learns the dependencies between substructures through Masked Substructure Modeling (MSM), and strengthens the discriminability of representations by means of generative objectives. Despite the significant progress made by the aforementioned methods, most of them fail to consider the semantic guiding role of meta-paths, making it difficult to targetedly mine semantic correlations in heterogeneous data.

3. Problem Definition

In the following, we introduce some essential concepts and give a formal definition of heterogeneous graph learning.

Heterogeneous Graph. A heterogeneous graph refers to a graph structure encompassing multiple node types or various edge types. It is typically denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$, where $\mathcal{V}$ and $\mathcal{E}$ represent the node set and edge set, respectively. $\mathcal{T}_v$ stands for the set of node types, and $\mathcal{T}_e$ stands for the set of edge types. Specifically, $\mathcal{G}$ is classified as a heterogeneous graph when $|\mathcal{T}_v| \geq 2$ or $|\mathcal{T}_e| \geq 2$; otherwise, it is a homogeneous graph.

Meta-Path. A meta-path is a path with specific semantic meaning defined on a heterogeneous graph, characterized by an alternating sequence of node types and edge types. It is formally expressed as $P = T_{v_1} \xrightarrow{T_{e_1}} T_{v_2} \xrightarrow{T_{e_2}} \dots \xrightarrow{T_{e_{k-1}}} T_{v_k}$, where $T_{v_i} \in \mathcal{T}_v$ (i.e., T_{v_i} is a node type from the set $\mathcal{T}_v$) and $T_{e_j} \in \mathcal{T}_e$ (i.e., T_{e_j} is an edge type from the set $\mathcal{T}_e$).

Meta-Path Graph. A meta-path graph refers to a simplified graph constructed using predefined meta-paths P(i.e., paths with specific semantic meaning defined on heterogeneous graphs). It preserves all nodes from the original heterogeneous graph and establishes new edges exclusively between node pairs that conform to the semantic constraints of the given meta-paths.

Heterogeneous Graph Learning. Formally, let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$ represent a heterogeneous graph, where $\mathcal{V} = \mathcal{V}_{\text{labeled}} \cup \mathcal{V}_{\text{unlabeled}}$ ($\mathcal{V}_{\text{labeled}}$ and $\mathcal{V}_{\text{unlabeled}}$ represent the sets of labeled and unlabeled nodes, respectively). Let $\mathcal{Y} = \{y_1, y_2, \dots, y_C\}$ be the predefined category label set with C classes. $\mathcal{K} = \{1, 2, \dots, K\}$ denote the set of cluster indices with K predefined clusters. For node classification, SMCD aims to learn a mapping function $f : \mathcal{V} \to \mathcal{Y}$, which predicts the label $y_v \in \mathcal{Y}$ for each unlabeled node $v \in \mathcal{V}_{\text{unlabeled}}$. For node clustering, SMCD seeks to learn a partitioning function $\Pi : \mathcal{V} \to \mathcal{K}$, which assigns each node $v \in \mathcal{V}$ to a unique cluster $\Pi(v) \in \mathcal{K}$.

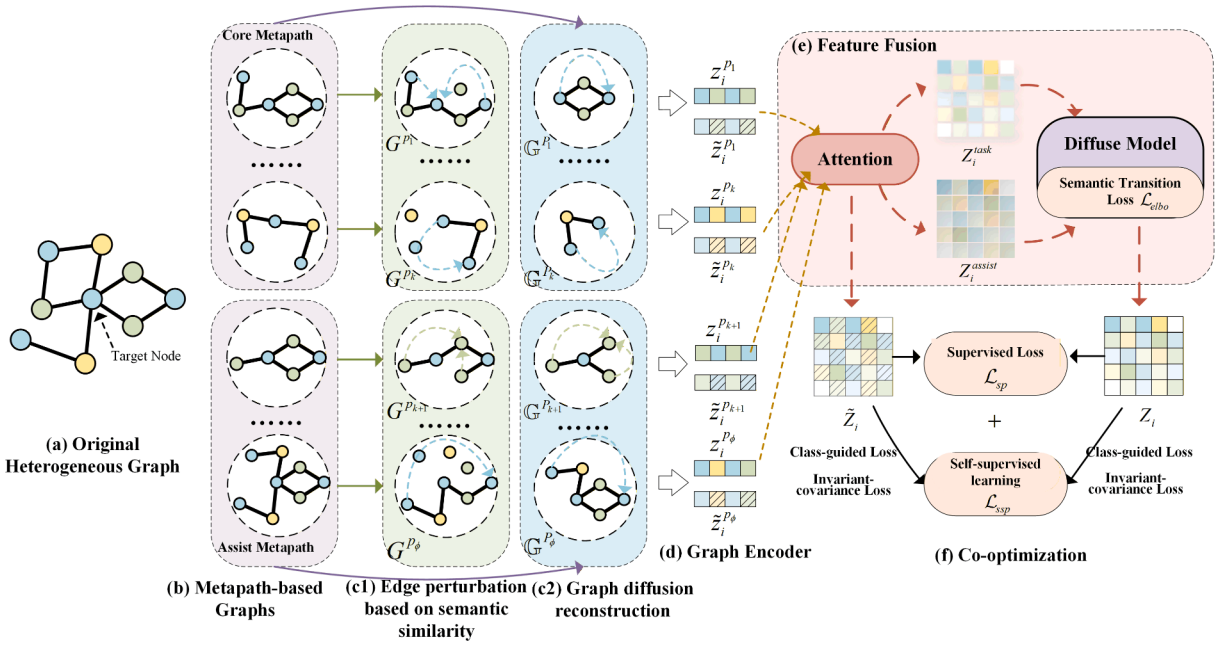


Fig. 1. Overall Architecture of the SMCD Model. Taking the original heterogeneous graph as input, the framework first constructs meta-path subgraphs based on core meta-paths and auxiliary meta-paths. Subsequently, it generates augmented views through two approaches: edge perturbation based on semantic similarity and graph diffusion reconstruction. Next, the graph encoder outputs multiple sets of node embeddings. Then, it fuses the embeddings of each meta-path graph via an attention mechanism to generate semantic-level views and task-level views, with the introduction of semantic transfer loss for representation optimization. Finally, it adopts a collaborative training approach that combines supervised loss and self-supervised loss.

4. Methods

This section elaborates on the technical design of the SMCD model, whose overall architecture is illustrated in Fig. 1. First, based on the original meta-paths, a suite of auxiliary meta-paths is devised to strengthen the core meta-paths. Then, two types of enhanced views are constructed using the core meta-paths and auxiliary meta-paths: For semantic-level enhancement, edge perturbation based on semantic similarity is performed on all meta-path graphs, and the semantics of the core meta-paths are enhanced with the semantics of the auxiliary meta-paths via a diffusion model. For task-level enhancement, the diffusion model is leveraged to select the top-k nodes with the most prominent semantic relevance for each node within the core meta-path graph, aiming to construct local neighborhoods and adjust the original graph structure. The model then adopts a two-stage aggregated graph encoder to obtain the final embeddings. Finally, a collaborative optimization learning objective combining self-supervision and supervision is utilized.

4.1. Graph Structure Transformation

The main methods of graph augmentation include feature transformation and graph structure transformation. Feature transformation focuses primarily on the attribute features of nodes or edges, which tends to overlook the semantic information embedded in the topological structure. In contrast, graph structure transformation, by reasonably adjusting the graph's topological structure, can not only effectively preserve core semantic information but also reduce the model's over-reliance on specific node connections or local structures. Therefore, this paper adopts the transformation strategy of the graph structure, performing targeted transformations on the input graph data based on the adjacency matrix A of the original graph to produce two types of differentiated new views.

4.1.1. Meta-Path Design

We design new meta-paths as auxiliary meta-paths to augment the semantic information of task-specific meta-paths. The selection criteria for auxiliary meta-paths are as follows:

Semantic Association Anchoring Criterion. Auxiliary meta-paths are not required to share the same start/end node types as the core meta-path, but they must satisfy the dual constraints of target node semantic anchoring and semantic complementary mapping. These constraints do not rely on domain-specific prior knowledge and are applicable to all heterogeneous graph scenarios:

Target Node Anchoring: Auxiliary meta-paths must include the target node type of the core meta-path (e.g., the target node type is Paper (P) in the ACM dataset, which is contained in all auxiliary meta-paths including $\mathcal{APA}$, $\mathcal{SPS}$, and $\mathcal{PSPAP}$), ensuring that the semantics revolve around the task core.

Semantic Complementary Mapping: Auxiliary meta-paths and the core meta-path must share at least one edge type (e.g., $\mathcal{APA}$ and $\mathcal{PAP}$ share the P-A edge, while $\mathcal{SPS}$ and $\mathcal{PSP}$ share the P-S edge in the ACM dataset). Overlapping relation types guarantee semantic relevance while avoiding complete semantic redundancy.

Regardless of the domain, the core value of auxiliary meta-paths selected according to this criterion lies in complementing rather than deviating from the core semantics through target node anchoring and relation sharing.

Path Length Adaptation Criterion. To balance semantic expressiveness and computational feasibility, a unified length constraint rule is established, which is applicable to heterogeneous graphs across all domains:

Non-homologous auxiliary meta-paths (with different start/end node types from the core meta-path): The length should be consistent with that of the core meta-path, ensuring semantic transmission efficiency and avoiding semantic attenuation caused by excessively long paths in cross-domain scenarios.

High-order homologous auxiliary meta-paths (with the same start/end node types as the core meta-path): The length should be ≤

the length of the core meta-path. This enriches semantic dimensions while controlling computational costs in cross-domain scenarios, thus preventing efficiency issues arising from differences in dataset scale.

Taking the ACM dataset as an example, this heterogeneous graph comprises three node types (Author, Paper, and Subject) and two edge types (Paper-Author and Paper-Subject), with the target node type being Paper(P). Accordingly, we select $\mathcal{PAP}$ and $\mathcal{PSP}$ as the task-level meta-paths. Furthermore, we design three new meta-paths-$\mathcal{APA}$, $\mathcal{SPS}$, and $\mathcal{PSPAP}$-as auxiliary meta-paths.

4.1.2. Edge Perturbation

In heterogeneous graphs, different meta-paths represent distinct semantic associations[18]. If multiple meta-path instances exist between two nodes, it indicates that the two nodes have a strong association and high semantic similarity. To fully mine the semantic and structural information of the heterogeneous graph $\mathcal{G}$, this paper adopts the semantic similarity-based edge deletion strategy to perturb graph edges, thereby eliminating edges with weak associations. Specifically, given a set of predefined meta-paths $\Phi = \{P_1, P_2, \ldots, P_\phi\}$ (where ϕ represents the total count of meta-paths and P_k denotes the k-th meta-path). This paper leverages the core idea of PathSim [48] similarity. It calculates the semantic weight $w_{uv}^{P_k}$ between any two nodes u and v in the heterogeneous graph $\mathcal{G}$ based on meta-path P_k using Formula 1:

$$w_{uv}^{P_k} = \left| \{ p_{u \to v} \mid p_{u \to v} \in P_{uv}^{P_k} \} \right| \tag{1}$$

Here, $p_{u\to v}$ represents the set of path instances from node u to node v that conform to meta-path P_k. We can construct a semantic weight-enhanced meta-path graph $\mathcal{G}^{p_k} = (\mathcal{V}, A^{P_k}, W^{P_k})$, where $\mathcal{V}$ is the node set, A^{P_k} is the adjacency matrix, and W^{P_k} is the semantic weight matrix.

To achieve effective perturbation, this paper introduces a masking ratio parameter ϑ ($\vartheta \in (0, 1)$). All edges within the meta-path graph $\mathcal{G}^{p_k}$ are ordered in ascending sequence according to their semantic weights w_{uv}^{k}, and the bottom $\vartheta \times 100\%$ of edges with low weights are masked (i.e., temporarily deleted). The damaged meta-path graph obtained after the masking operation is denoted as G^{p_k}.

4.1.3. Graph Diffusion

To highlight the strong semantic correlations between nodes and reduce the interference of redundant connections on embedding learning, this study adopts a general graph diffusion strategy. Specifically, the strategy is applied to the meta-path graph $\mathcal{G}^{p_k}$ with semantic weights, aiming to generate a task-relevant meta-path subgraph $\mathbb{G}^{P_k}$. During the diffusion process, noise is introduced to perturb the original relationships of the meta-path graph, as specifically shown in Formula 2:

$$q(x_t|x_0) = N(x_t; \sqrt{\bar{a}_t}x_0, (1-\bar{a}_t)\mathbf{I}), \quad \bar{a}_t = \prod_{t'=1}^{t}(1-\beta_{t'}) \tag{2}$$

where $t \in \{1 \ldots T\}$ denotes the diffusion step, N represents the Gaussian distribution, and $\mathbf{I}$ is the identity matrix. The parameter $\beta_t \in (0, 1)$ is used to adjust the intensity of Gaussian noise imposed in the t-th diffusion step. The initial state is $x_0 = S^{P_k}$, where the element $S_{ij}^{P_k}$ characterizes the semantic correlation degree between node i and node j, calculated as shown in Formula 3:

$$S^{p_k} = D^{-1/2}\left(W^{P_k} + I_N\right)D^{-1/2} \tag{3}$$

Here, I_N is the identity matrix; D is a diagonal matrix with $D_{ii} = \sum_{j=1}^{N}(W_{ij}^{P_k} + I_{N,ij})$. When $t \to \infty$, the state x_T ultimately converges to the standard Gaussian distribution.

In the reverse process, the core task is to iteratively recover the initial semantic correlation representation x_0 from the fully Gaussian-noised state x_T. The diffusion model learns to eliminate the added noise through a neural network, thereby recovering x_{t-1} from x_t, as shown in Formula 4:

$$p_\theta(x_{t-1}|x_t) = N(x_{t-1}; u_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{4}$$

where $u_\theta(x_t, t)$ and $\Sigma_\theta(x_t, t)$ denote the mean and covariance of the Gaussian distribution, respectively, which can be obtained by a neural network with parameters θ. To calculate the noise introduced in the t-th diffusion step, the mean u_θ is reparameterized as shown in Formula 5:

$$u_\theta(x_t, t) = \frac{1}{\sqrt{\bar{a}_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{a}_t}}\epsilon_\theta(x_t, t)) \tag{5}$$

Here, $\epsilon_\theta(x_t, t)$ is implemented by a Multi-Layer Perceptron (MLP), aiming to compute the predicted value $\hat{x}_0$ based on the input x_t and the corresponding time step t. After obtaining $\hat{x}_0$, the top-k adjacent nodes with the highest semantic correlation degree are selected for each node to construct local neighborhoods, and the original graph structure is adjusted to obtain the meta-path subgraph $\mathbb{G}^{P_k}$ for node embedding learning, as specifically shown in Formula 6:

$$\widetilde{A} = top_{rank}(\hat{x}_0(i, :), k) \tag{6}$$

where k regulates the subgraph size through the number of neighbors of a node, and the function top_{rank} returns the indices of nodes sorted by their semantic similarity.

4.2. Network Architecture

4.2.1. Feature Preprocessing

Since the input feature dimensions of nodes with varying types may differ, we map the features of any node of type Λ (where $\Lambda \in \mathcal{T}_v$; $\mathcal{T}_v$ is the set of all node types) to a shared feature space using Formula 7.

$$h_i = \sigma(W_\Lambda \cdot x_i + b_\Lambda) \tag{7}$$

where$x_i \in R^{d_\Lambda}$is the original feature of node v_i, with $h_i \in R^d$ being its projected feature. $W_\Lambda \in R^{d \times d_\Lambda}$ stands for the trainable mapping matrix for type Λ, $b_\Lambda \in R^d$ denotes the trainable bias vector, and $\sigma(\cdot)$ denotes an activation function.

4.2.2. Encoder

After obtaining different meta-path graphs through graph structure transformation, we use an encoder to acquire the representation of each meta-path graph. To this end, we introduce a graph attention layer [49] as the encoder, which is used to aggregate the neighbor embedding information of nodes within the meta-path graph. The key advantage of this layer lies in its reliance on the self-attention mechanism, which can automatically allocate differentiated weights based on the contribution of neighbor nodes to the target node. This effectively distinguishes the importance differences among different neighbors, and further generates exclusive embedding representations of nodes under the corresponding meta-path semantics by accurately aggregating neighbor features.

Specifically, for node v_i, the method for calculating the importance of its neighbor node v_j under a specific meta-path association is as Formula 8:

$$e_{ij}^{p_k} = LeakyReLU(m^{p_k T} \cdot [h_i \oplus h_j]) \tag{8}$$

Here, $m^{p_k} \in \mathbb{R}^{2d}$ denotes the intra-graph attention vector corresponding to meta-path P_k (where d is the node embedding dimension), and the symbol $\oplus$ represents the vector concatenation operation. After calculating the importance scores of all meta-path-associated neighbor nodes of the target node v_i, we normalize these scores to obtain the weight coefficients $\alpha_{i,j}^{p_k}$ for each neighbor node using Formula 9. Finally, we integrate the features of all neighbor nodes via weighted summation to derive the embedding representation of node v_i under the semantics of this meta-path using Formula 10.

$$\alpha_{i,j}^{p_k} = \frac{\exp\left(e_{i,j}^{p_k}\right)}{\sum_{s \in \mathcal{N}_i^{p_k}} \exp\left(e_{i,s}^{p_k}\right)} \tag{9}$$

$$z_i^{p_k} = PRELU\left(\sum_{j\in\mathcal{N}_i^{p_k}} \alpha_{i,j}^{p_k}\cdot h_j + b^{p_k}\right) \tag{10}$$

Here, $b^{p_k}\in\mathbb{R}^d$ denotes the bias vector dedicated to aggregation under metapath p_k, and $PRELU(\cdot)$ denotes an activation function.

To stabilize the learning process of intra-graph attention and enrich the information dimension of embedding representations, we extend the attention mechanism to multi-head attention. Concretely, we execute M independent attention mechanisms in parallel and concatenate the embeddings output by these mechanisms to obtain the following representation using Formula 11.

$$z_i^{p_k} = \bigoplus_{m=1}^{M}\sigma\left(\sum_{j\in\mathcal{N}_i^{p_k}} [\alpha_{i,j}^{p_k}]_k\cdot \mathbf{W}_m h_j + b^{p_k}\right) \tag{11}$$

where $z_i^{p_k}$ represents the trainable embedding of node v_i within the meta-path graph G^{p_k}. For a given collection of meta-path graphs sets $\{G^{p_1},\ldots,G^{p_\phi},\mathbb{G}^{p_1},\ldots,\mathbb{G}^{p_\phi}\}$, once node features are input into the neighborhood-level attention module, we derive 2ϕ groups of node embeddings specific to each meta-path graph, denoted as $\{z_i^{p_1},\ldots,z_i^{p_\phi},\widetilde{z}_i^{p_1},\ldots,\widetilde{z}_i^{p_\phi}\}$.

4.2.3. Feature Fusion

After obtaining node embeddings within each meta-path graph, we need to fuse these embeddings to derive the final node embeddings. We employ a graph-level attention mechanism to automatically capture the importance of different meta-paths and use the learned weights to fuse the corresponding node embeddings.

For the task-level view, we fuse the node embeddings corresponding to task-level meta-paths, i.e., $\{\widetilde{z}_i^{p_1},\ldots,\widetilde{z}_i^{p_{\text{task}}}\}$, to obtain the final node embedding $\widetilde{\mathbb{Z}}_i$. Specifically, we calculate the average of all Author-type nodes in the specific meta-path graph $\mathbb{G}^{p_k}$ to derive the summary vector of $\mathbb{G}^{p_k}$. Subsequently, we employ a learnable vector $\mathbf{q}_A\in\mathbb{R}^d$ to calculate the importance of each meta-path graph. The formula is given by Formula 12:

$$\mathbf{e}^{p_k} = q_A^T\cdot\frac{1}{|\mathcal{V}_A|}\sum_{i\in\mathcal{V}_A}\tanh\left(\mathbf{W}_A\cdot\mathbf{z}_i^{p_k}+\mathbf{b}_A\right) \tag{12}$$

Here, $\mathbf{W}_A\in\mathbb{R}^{d\times d}$ is a learnable weight matrix, and $\mathbf{b}_A\in\mathbb{R}^d$ denotes a learnable bias vector. Finally, we normalize the importance scores using the softmax function to obtain weight coefficients. Subsequently, we aggregate the node embeddings from all task-level meta-path graphs for each node to derive the final node embedding. The relevant formulas are given by Formula 13 and 14:

$$\beta^{p_k} = softmax(e^{p_k}) \tag{13}$$

$$\widetilde{\mathbb{Z}}_i = \sum_{k=1}^{task}\beta^{p_k}\cdot z_i^{p_k} \tag{14}$$

For the semantic-level contrastive view, we first fuse the node embeddings corresponding to task-level meta-paths $\{z_i^{p_1},\ldots,z_i^{p_{\text{task}}}\}$ and those corresponding to auxiliary-level meta-paths $\{z_i^{p_1},\ldots,z_i^{p_{\text{assist}}}\}$ using the same method, respectively, to obtain the final task-level node embedding Z_i^{task} and the final auxiliary-level node embedding Z_i^{assist}. Subsequently, we enhance the task-level node embeddings using the auxiliary-level node embeddings to obtain the final node embeddings $\mathbf{Z}_i$. Specifically, we treat the auxiliary-level node embeddings as diffusion sources, use the task-level node embeddings to provide denoising training signals, and construct this view through a diffusion model, thereby enhancing the model's robustness to data noise. The relevant formulas are as Formula 2,4,5. Among them, the initial state is $x_0 = Z_{diff}$, and its calculation method is shown in Formula 15:

$$Z_{diff} = Z^{\text{task}} + \lambda\cdot Z^{\text{assist}} \tag{15}$$

where the hyperparameter $\lambda\in[0,1]$ balances the contribution of embeddings at the task and auxiliary levels, ensuring that the diffusion process starts with a semantically integrated initial state. Finally, Equation 16 is used to evaluate the diffusion loss in the process of generating the contrastive view.

$$L_{elbo} = E_{t\sim U(1,T)}E_{(Z^{\text{task}})}[||\epsilon_\theta(x_t,t)-Z^{\text{task}}||_2^2] \tag{16}$$

where $||\epsilon_\theta(x_t,t)-Z^{\text{task}}||_2^2$ is a semantic transition loss term. Through this method, the auxiliary meta-path graph embeddings Z_i^{assist} are converted into the target semantic space by the diffusion model. This transformation process enables accurate modeling of the semantic conversion from auxiliary meta-paths to task meta-paths, thereby effectively extracting task-relevant information to generate semantic-level contrastive views.

4.3. Co-optimization

This paper adopts a training paradigm of co-optimization between self-supervised learning and supervised learning, flexibly adjusting the learning strategy according to the characteristics of label distribution in the dataset: For labeled nodes, supervised learning is used to guide embedding learning with labels, ensuring the model captures task-relevant discriminative features; for unlabeled nodes, self-supervised learning is leveraged to explore the intrinsic structure and semantic correlations of the data itself, fully utilizing the value of unlabeled data. These two learning methods complement each other and are co-optimized, which not only ensures the supervisory guidance of the labeled data but also maximizes the potential information of the unlabeled data, effectively improving the model's generalization capacity and the efficiency of data utilization.

Supervised Loss. To enable the model to acquire the embeddings of discriminative nodes, we constructed the supervised loss by minimizing the cross-entropy between the true labels of all labeled nodes and the predicted results of the model. The specific loss formula is as Formula 17:

$$\mathcal{L}_{sp} = -\sum_{i\in\mathcal{Y}}\mathbf{Y}_i\ln\left(\text{MLP}(\mathbf{Z}_i^F)\right) \tag{17}$$

where MLP denotes the multi-layer perceptron (prediction head) for classification tasks, $\mathcal{Y}$ denotes the index set of labeled nodes, and $\mathbf{Y}_i$ and $\mathbf{Z}_i^F$ correspond to the label and multi-view fused embedding of labeled node i, respectively. Guided by the supervision of labeled data, the model iteratively optimizes all learnable parameters through backpropagation, allowing node embeddings to accurately capture task-relevant discriminative features and thereby improve the final prediction performance.

Self-supervised Learning Loss. The self-supervised learning loss $\mathcal{L}_{\text{ssp}}$ consists of two components: invariant-covariance loss and class-guided loss. For the invariant-covariance loss, the core objective of this paper is to enhance the correlation of features in the same dimension across different views while reducing redundant correlations among features of different dimensions within the same view, thereby optimizing the structural expression ability of embeddings.

First, we perform instance-wise standardization on node feature embeddings to make the distribution of each feature dimension satisfy a mean of 0 and a standard deviation of 1. The formula is as Formula 18.

$$\hat{\mathbf{Z}} = \frac{\mathbf{Z}-\mu(\mathbf{Z})}{\sigma(\mathbf{Z})} \tag{18}$$

Subsequently, based on the standardized embeddings $\hat{\mathbf{Z}}$ and $\hat{\mathbb{Z}}$, we obtain the complete expression of the invariant-covariance loss using Formula 19.

$$\mathcal{L}_{ic} = \frac{1}{\sum_i \text{Diag}(\hat{\mathbf{Z}}^T\hat{\mathbb{Z}})_i - \theta_1\left(\sum_{i\neq j}\text{OFFDiag}(\hat{\mathbf{Z}}^T\hat{\mathbf{Z}})_{ij}^2 + \sum_{i\neq j}\text{OFFDiag}(\hat{\mathbb{Z}}^T\hat{\mathbb{Z}})_{ij}^2\right)} \tag{19}$$

Here, $\sum_i \text{Diag}(\hat{\mathbf{Z}}^T\hat{\mathbb{Z}})_i$ is the invariance objective function, which serves to strengthen the feature correlation of the same dimension across

views; $\sum_{i\neq j}\text{OFFDiag}(\hat{\mathbf{Z}}^T\hat{\mathbf{Z}})_{ij}^2$ and $\sum_{i\neq j}\text{OFFDiag}(\hat{\mathbb{Z}}^T\hat{\mathbb{Z}})_{ij}^2$ are covariance objective functions, designed to achieve decorrelation between different dimensions within the same view. $\text{Diag}(\cdot)_i$ represents extracting the i-th diagonal element of the matrix. $\text{OFFDiag}(\cdot)_{ij}$ $(i \neq j)$ denotes extracting the off-diagonal element of the matrix. and $\theta_1 > 0$ is a balancing hyperparameter that adjusts the strength of the covariance constraint, enabling the optimal synergy between the two objectives of invariance enhancement and dimension decorrelation.

Class-guided loss specifically regulates the distance distribution between feature embeddings of nodes of different classes. We employ the feature embeddings of labeled data to implement class constraints, making features of the same class more similar and features of different classes more distinct. Let C be the number of classes in the labeled data, and define the sample set $\mathcal{Y} = \{y_1, y_2, \ldots, y_C, y_0\}$ (where y_0 represents unlabeled data). The distance between two sample sets y_p and y_q is defined as Formula 20.

$$\text{dis}(\hat{Z}_{y_p}, \hat{Z}_{y_q}) = 1 - \frac{1}{|y_p||y_q|}\sum_{i\in y_p, j\in y_q}[\hat{Z}_i\hat{Z}_j^T]_{i,j} \tag{20}$$

where i and j are samples in the sets, respectively, and $\hat{Z}_i\hat{Z}_j^T$ is used to gauge the similarity between the embeddings of the sample.

$$s(\hat{Z}) = \frac{1}{C}\sum_{i=1}^{C}\text{dis}(\hat{Z}_{y_i}, \hat{Z}_{y_i}) \tag{21}$$

$$d(\hat{Z}) = \frac{1}{C(C-1)}\sum_{i\neq j}\text{dis}(\hat{Z}_{y_i}, \hat{Z}_{y_j}) \tag{22}$$

$$u(\hat{Z}) = \frac{1}{C}\sum_{i=1}^{C}\text{dis}(\hat{Z}_{y_i}, \hat{Z}_{y_0}) \tag{23}$$

Here, $s(\hat{Z})$ represents the average intra-class distance, $d(\hat{Z})$ denotes the average inter-class distance across different classes, and $u(\hat{Z})$ stands for the average distance between unlabeled and labeled classes.

For the standardized embeddings $\hat{\mathbf{Z}}$ and $\hat{\mathbb{Z}}$, the class-guided loss is defined as Formula 24.

$$\mathcal{L}_{cg} = \left(\frac{s(\hat{\mathbf{Z}})}{d(\hat{\mathbf{Z}})} + \frac{s(\hat{\mathbb{Z}})}{d(\hat{\mathbb{Z}})}\right) + \theta_2\left(u(\hat{\mathbf{Z}}) + u(\hat{\mathbb{Z}})\right) \tag{24}$$

where θ_2 is a balancing hyperparameter that adjusts the importance of "the sum of average distances between unlabeled and labeled classes," forming a synergistic constraint with the preceding term "the sum of intra-class/inter-class distance ratios." Combined with the invariant-covariance loss, the total self-supervised loss is defined as Formula 25.

$$\mathcal{L}_{ssp} = \delta\mathcal{L}_{ic} + (1-\delta)\mathcal{L}_{cg} \tag{25}$$

where δ is a balancing coefficient with a value range of $\delta \in [0, 1]$, which is used to adjust the weight ratio between the invariant-covariance loss $\mathcal{L}_{ic}$ and the class-guided loss $\mathcal{L}_{cg}$ in the total self-supervised loss $\mathcal{L}_{ssp}$.

The overall loss function is defined as Formula 26.

$$\mathcal{L} = \mathcal{L}_{sp} + \mathcal{L}_{ssp} + \mathcal{L}_{elbo} \tag{26}$$

Time Complexity Analysis. We define N as the total number of nodes in the heterogeneous graph, Φ as the total number of meta-paths (including core and auxiliary meta-paths), E_k as the number of edges in the k-th meta-path graph, T as the number of diffusion steps, d as the embedding dimension, h as the number of heads in multi-head attention, and q as the number of top semantically relevant nodes selected during local neighborhood reconstruction. Since Φ is a constant, the time complexity of semantic weight calculation is $O(E_k)$; the diffusion process has a time complexity of $O(T \times E_k \times d^2)$; the reconstruction of the graph structure incurs $O(N^2 \log N)$; the attention encoding of the multi-head graph intra-meta-path operates at $O(h \times (N \times d^2 + E_k \times d))$; and the fusion of graph-level attention from cross-meta-path has a time complexity of $O(N \times d^2)$. Thus, the total time complexity of the diffusion model for reconstructing meta-path subgraphs is $O(T \times E_k \times d^2 + N^2 \log N)$. In particular, the term $O(N^2 \log N)$ can be further reduced to $O(Nq \log q)$ through local neighborhood constraints, adapting to the computational power limit of ultra-large-scale graphs. For the diffusion model used to generate semantic-level contrastive views, its time complexity is $O(T \times E_k \times d^2 + N \times d^2)$.

Table 1
Overview of Datasets.

Datasets	Node	Relation	Metapath(core)	Metapath(assist)
ACM	**P**paper:3025 **A**uthor:5912 **S**ubject:57	P-A:9936 P-S:3025	PAP PSP	APA SPS PSPAP
DBLP	**P**aper:14328 **A**uthor:4057 **C**onference:20	P-A:19645 P-C:14328	APA APCPA	PAP
IMDB	**M**ovie:4664 **A**ctor:5841 **D**irect:2270	M-A:13983 M-D:4661	MAM MDM	AMA DMD MAMDM

Space Complexity Analysis. The semantic weight matrix has a space complexity of $O(E_k)$; the diffusion process requires $O(E_k \times T \times d)$; graph structure reconstruction demands $O(N)$; the multi-head graph attention encoder in the intra-meta-path occupies $O(h \times d^2)$; and graph-level attention fusion from cross-meta-path uses $O(d^2)$. The total space complexity of the diffusion model for reconstructing meta-path subgraphs is $O(E_k \times T \times d)$, while the space complexity of the diffusion model for generating semantic-level contrastive views is $O(E_k \times T \times d)$.

5. Experimental Procedure and Results

In this section, we carry out comprehensive experiments on three real-world datasets to verify the effectiveness and efficiency of the SMCD model.

5.1. Datasets and baseline models

Datasets. We selected three publicly accessible real-world heterogeneous graph datasets for the experiments. Detailed statistics are presented in Table 1.

- **ACM** [50]: The target nodes correspond to papers, categorized into three classes: database, data mining, and wireless communication. Each paper is associated with an average of 3.28 authors and one subject.
- **DBLP** [48]: The target nodes are authors, classified into four categories: database, data mining, machine learning, and information retrieval. On average, each author has 4.84 papers.
- **IMDB** [50]: The target nodes are movies, grouped into three genres: action, comedy, and drama. Each movie is associated with an average of 3 actors and one director.

Baselines. We select diverse baselines across two application domains to evaluate the performance of our SMCD model, and the baselines for node classification are displayed below.

- **GCN** [51]: It proposes a Graph Convolutional Network (GCN) based on the first-order approximation of spectral graph convolution. Through the normalized adjacency matrix and inter-layer propagation rules, it enables efficient semi-supervised node classification on graph-structured data.
- **HAN** [25]: HAN is a semi-supervised node classification method for heterogeneous graphs. It aggregates node features and structural information through a meta-path guided attention mechanism to achieve effective classification.

Table 2
Evaluation of Node Classification Performance Against Baseline Methods.

Datasets	ACM		DBLP		IMDB	
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
GCN	91.68 ± 0.52	91.65 ± 0.51	87.35 ± 1.35	88.56 ± 1.31	56.14 ± 1.66	63.47 ± 0.88
HAN	90.96 ± 0.76	90.86 ± 0.76	92.96 ± 0.32	93.43 ± 0.31	60.07 ± 1.40	65.92 ± 0.73
MAGNN	90.87 ± 1.16	90.77 ± 1.15	93.19 ± 0.52	93.81 ± 0.49	55.73 ± 2.51	64.21 ± 1.43
Simple-HGN	93.27 ± 0.49	93.19 ± 0.50	93.57 ± 0.52	94.06 ± 0.48	63.97 ± 1.40	67.71 ± 0.82
HGT	90.33 ± 0.70	90.27 ± 0.68	92.28 ± 1.04	93.00 ± 0.85	62.13 ± 1.60	66.90 ± 0.84
SeHGNN	94.05 ± 0.35	93.98 ± 0.36	94.92 ± 0.38	95.28 ± 0.36	66.72 ± 0.41	68.43 ± 0.39
RpHGNN	92.47 ± 0.97	92.37 ± 1.00	94.90 ± 0.25	95.26 ± 0.23	66.97 ± 0.64	69.16 ± 0.61
HAGNN	92.54 ± 0.55	92.47 ± 0.56	94.41 ± 0.28	94.80 ± 0.26	64.43 ± 1.21	67.83 ± 0.65
DiffGraph	90.75 ± 0.69	90.2 ± 0.74	91.81 ± 0.25	92.41 ± 0.24	62.8 ± 0.90	68.39 ± 4.51
HGCMA	92.02 ± 0.02	91.94 ± 0.06	93.38 ± 0.03	93.77 ± 0.03	57.77 ± 0.30	63.35 ± 0.34
CALHG	94.49 ± 0.18	94.49 ± 0.18	95.17 ± 0.35	95.55 ± 0.32	66.54 ± 0.39	68.65 ± 0.35
SMCD	**96.30 ± 0.01**	**96.20 ± 0.11**	**97.18 ± 0.04**	**97.42 ± 0.04**	**68.59 ± 0.38**	**75.06 ± 0.09**

Table 3
Evaluation of Node Clustering Performance Against Baseline Methods.

Datasets	ACM		DBLP		IMDB	
	NMI	ARI	NMI	ARI	NMI	ARI
DGI	55.74	58.79	16.55	11.22	1.50	-0.23
GraphCL	21.66	17.31	46.49	47.19	2.01s	0.68
DMGI	53.29	50.39	68.26	74.86	7.14	4.62
HeCo	60.58	63.14	71.92	77.38	4.50	4.28
HGCML	53.58	51.92	71.18	76.94	9.24	6.70
HGCMA	69.95	74.40	76.61	81.45	10.29	9.44
HAGNN	80.43	81.57	83.51	88.51	16.44	13.05
SMCD	**85.89**	**88.37**	**89.48**	**92.85**	**21.32**	**19.38**

- **MAGNN** [9]: It encodes meta-path instances and intermediate node information through intra-meta-path aggregation with an attention mechanism. It generates node representations by combining node content transformation and inter-meta-path aggregation.
- **Simple-HGN** [8]: It unifies the feature space via type-specific linear transformations, integrates residual connections, multi-head attention, and L_2 normalization, and achieves excellent performance on the node classification task of the HGB benchmark without relying on meta-paths.
- **HGT** [27]: It characterizes heterogeneous attention by designing node and edge type-dependent parameters, and achieves efficient and scalable training by integrating the heterogeneous mini-batch graph sampling algorithm (HGSampling).
- **SeHGNN** [13]: It precomputes neighbor features through mean aggregation, expands the receptive field using single-layer long meta-paths, and fuses multi-meta-path information via a Transformer-based semantic fusion module, achieving both high accuracy and fast training speed.
- **RpHGNN** [52]: It adopts a propagation-update iteration framework, controls complexity through random projection compression, and reduces information loss by combining relation-level neighbor collection and parity propagation schemes, achieving both high performance and fast speed across multiple datasets.
- **HAGNN** [10]: It combines meta-path neighbors and direct neighbors with a two-stage framework of "meta-path-driven intra-type aggregation + meta-path-free inter-type aggregation" to fully explore the heterogeneity of heterogeneous graphs.
- **DiffGraph** [53]: This approach transforms auxiliary heterogeneous data into the target semantic space, allowing for the precise extraction of task-relevant information from multi-behavior data.
- **HGCMA** [17]: By constructing multiple views and performing cross-view contrastive learning, it fuses topological and semantic information to learn robust node representations, thereby improving the performance of tasks such as node classification.
- **CALHG** [20]: CALHG is a contrastive auxiliary learning model for heterogeneous graphs. It encodes rich information of heterogeneous graphs into low-dimensional node embeddings to improve the performance of related tasks.

We adopt HGCMA [17], along with the following ap approaches, as baselines for the node clustering task.

- **DGI** [35]: Its core lies in learning by maximizing the mutual information between subgraph patch representations generated by GCN and high-order graph summaries, without relying on random walks.
- **GraphCL** [54]: GraphCL designs four graph augmentation methods and systematically investigates their combined effects. It can generate high-quality graph representations without tuning augmentation intensity or relying on complex GNN architectures.
- **DMGI** [42]: DMGI fuses multi-relational node embeddings through consistency regularization, distinguishes real samples across types via a general discriminator, and infers relation importance with an attention mechanism.
- **HeCo** [40]: HeCo constructs dual views to capture diverse structural information. It then employs cross-view contrastive learning to enable the two views to collaboratively supervise the extraction of positive and negative embeddings, ultimately learning high-quality node embeddings.
- **HGCML** [41]: It generates multiple views based on meta-paths, models view correlations through intra/inter-meta-path contrastive learning, and mitigates sampling bias via topological and semantic positive sampling to learn discriminative node representations.

Parameter settings. In the experiments, the hyperparameter controlling the contribution of auxiliary meta-paths is set to $\lambda = 0.8$, and the meta-path graph masking ratio ϑ is determined through searching from the set $\{0.1, 0.3, 0.5, 0.7, 0.9\}$. The learning rate is set to 0.001, the hyperparameter δ regulating the contribution of invariance-covariance loss in the self-supervised loss is $\delta = 0.5$, the number of diffusion steps $T = 50$, the embedding dimension is set to 64, and the model parameters $\theta_1 = 0.5$ and $\theta_2 = 0.5$. For the node classification task, Macro-F1 and Micro-F1 are employed to assess the performance of the SMCD model; for the node clustering task, NMI and ARI serve as evaluation metrics. To ensure fair comparison, all parameters of baseline models are set to their default optimal values. For different datasets, their optimal parameters are searched separately, and each experiment is repeated 5 times. The average values and standard deviations of the metrics are reported. The dataset is split into training, validation, and test sets: 3% of labeled nodes are used for training, 29% for validation, and 68% for testing.

Experiments are conducted on the Ubuntu 20.04.6 operating system. The hardware environment consists of an Intel(R) Xeon(R) Gold

Table 4
Ablation Study on key components of SMCD.

Datasets	ACM		DBLP		IMDB	
Variants	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
SMCD_1	20.32 ± 0.31	35.99 ± 0.10	32.80 ± 0.53	45.89 ± 0.65	44.95 ± 0.14	60.19 ± 0.32
SMCD_2	90.57 ± 0.03	90.51 ± 0.03	92.76 ± 0.05	93.02 ± 0.04	68.12 ± 0.04	74.08 ± 0.03
SMCD_3	91.48 ± 0.78	93.53 ± 0.34	90.19 ± 0.14	91.10 ± 0.11	64.33 ± 0.11	70.82 ± 0.10
SMCD_4	89.57 ± 0.07	89.55 ± 0.08	92.05 ± 0.30	95.19 ± 0.37	67.45 ± 0.33	72.11 ± 0.50
SMCD	**96.30 ± 0.01**	**96.20 ± 0.11**	**97.18 ± 0.04**	**97.42 ± 0.04**	**68.59 ± 0.38**	**75.06 ± 0.09**

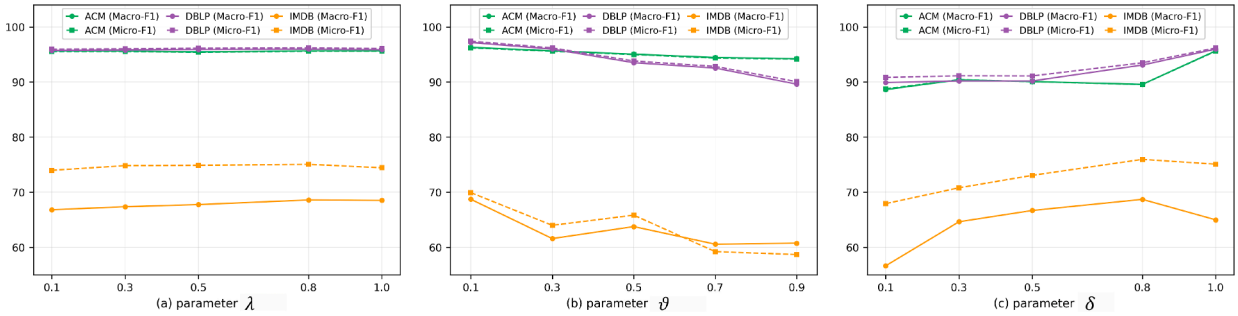


Fig. 2. Hyperparameter analysis for λ, ϑ, and δ.

5317 CPU (3.00GHz), 512GB memory, and a Tesla A100 80G GPU, with Python 3.9 as the software environment.

5.2. Evaluation on Node Classification

The node classification task aims to train a classification model using nodes with known category labels to accurately predict the labels of unlabeled nodes. As shown in Table 2, the experimental results clearly demonstrate that the SMCD model proposed in this paper exhibits superior performance in both accuracy and training efficiency. The core advantage of the SMCD model lies in its effective mining of the deep structural and semantic correlations of heterogeneous graphs via a semantic similarity-based edge perturbation strategy and enhancing the semantic expression of core meta-paths with the help of auxiliary meta-paths. This design enables it to achieve significantly improved performance compared to traditional meta-path-based methods, fully highlighting the unique advantage of the proposed method in fusing graph structural and semantic information. In addition, the model adopts a self-supervised and supervised collaborative optimization strategy that flexibly adapts to label distribution, which not only ensures node representations possess strong discriminability and good generality but also flexibly adapts to practical scenarios with different degrees of label scarcity, further enhancing the model's practicality and generalization ability.

5.3. Evaluation on Node Clustering

We adopt the K-means algorithm [55] to perform clustering on the learned embeddings of all labeled nodes, with the number of clusters for each dataset strictly set to the corresponding number of data categories. To mitigate experimental instability caused by the randomness of initial values, each experiment is independently repeated 5 times, and the final results are reported as the average values, as detailed in Table 3. The experimental results clearly demonstrate that the SMCD model consistently achieves optimal performance across all datasets, further verifying the effectiveness and superiority of its design. Meanwhile, it can be seen that approaches implementing edge perturbation based on semantic similarity generally outperform those with random edge deletion. This phenomenon reconfirms the core value of semantic similarity-based perturbation in deeply mining the semantic and structural information of heterogeneous graphs and improving the quality of representation learning.

Table 5
Robustness Analysis.

Noise Ratios	10%		30%		50%	
	Macro-F1	Micro-F1	Macro-F1	Micro-F1	Macro-F1	Micro-F1
HeCo	94.32%	95.48%	90.25%	89.79%	88.32%	88.79%
HGCMA	97.19%	97.16%	92.99%	92.90%	90.89%	90.81%
SMCD	**100.00%**	**100.00%**	**99.99%**	**99.98%**	**99.99%**	**99.98%**

5.4. Ablation Study

To assess the efficacy of the following core modules during this research phase, we conduct ablation experiments: supervised learning, self-supervised learning, edge perturbation augmentation, and diffusion reconstruction. To systematically evaluate the contribution of each module, we perform 4 groups of experiments and report the mean values of Macro-F1 and Micro-F1 metrics.

5.4.1. Effectiveness of Supervised Learning

This part assesses the influence of supervised learning on the SMCD model via ablation experiments. The specific operation is to remove the supervised loss component $\mathcal{L}_{sp}$ from the total loss function, yielding the variant model SMCD_1. As illustrated in Table 4, the experimental findings clearly confirm the effectiveness of supervised learning. The results highlight the key role of supervised learning in improving model performance, whose core function is to provide the model with clear class-oriented signals and enhance the accuracy of classification decisions.

5.4.2. Effectiveness of Self-supervised Learning

This part evaluates the influence of self-supervised learning on the SMCD model through ablation experiments. The specific operation involves removing the self-supervised loss component $\mathcal{L}_{ssp}$ from the total loss function, yielding the variant model SMCD_2. As illustrated in Table 4, the experimental results clearly verify the effectiveness of self-supervised learning. On the ACM dataset, compared with SMCD_2, SMCD achieves an average Macro-F1 improvement of 6.3% and an average Micro-F1 improvement of 6.28%; on the DBLP dataset, the average Macro-F1 and Micro-F1 increase by 4.93% and 4.73%, respectively; on the IMDB dataset, the average Macro-F1 and Micro-F1 of SMCD are 2.77% and 1.82% higher than those of SMCD_2. These results

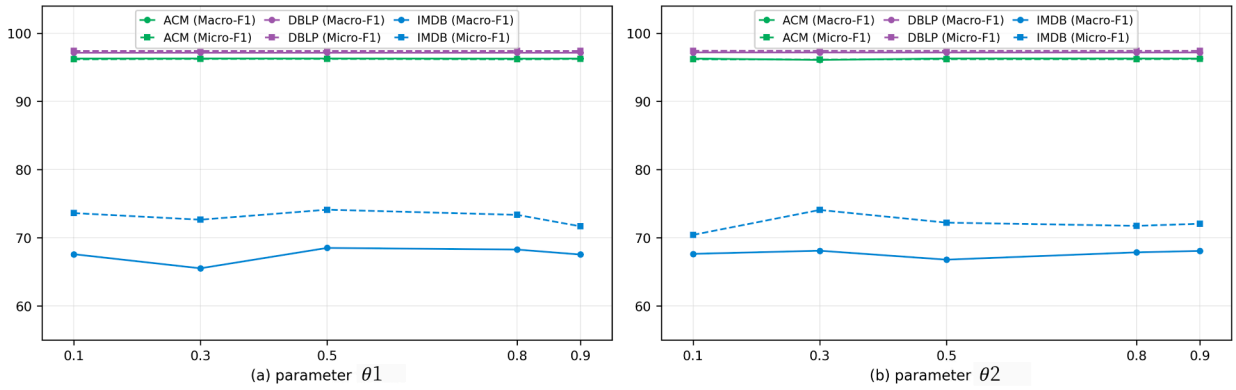


Fig. 3. Hyperparameter analysis for θ_1 and θ_2.

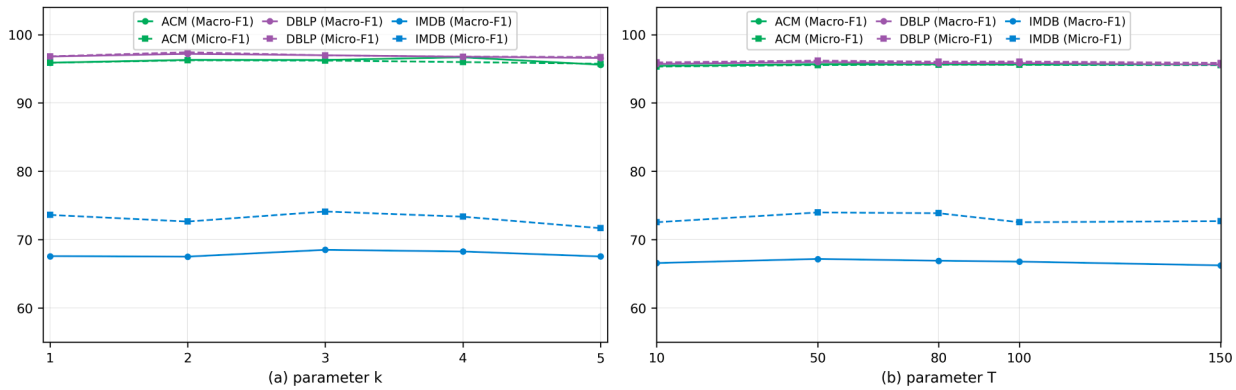


Fig. 4. Hyperparameter analysis for k and T.

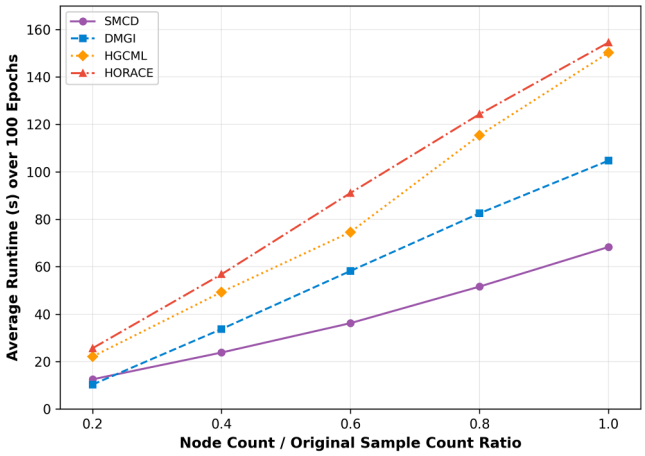


Fig. 5. Scalability Analysis.

underscore the key role of self-supervised learning in improving model performance, whose core function lies in mining the value of structural and semantic correlations through unlabeled data.

5.4.3. *Effectiveness of edge perturbation augmentation*

This section investigates the influence of edge perturbation augmentation on the overall performance of the SMCD framework by removing this module (denoted as SMCD_3). As illustrated in Table 4, the experimental findings demonstrate that SMCD consistently outperforms SMCD_3 across all datasets. On the ACM dataset, SMCD achieves an average Macro-F1 improvement of 5.26% and an average Micro-F1 improvement of 2.85% compared to SMCD_3; on the DBLP dataset, the average Macro-F1 and Micro-F1 increase by 7.75% and 6.94% respectively; on the IMDB dataset, the average Macro-F1 and Micro-F1 of SMCD are 8.83% and 6.51% higher than those of SMCD_3. These results confirm that by perturbing graph structural edges based on semantic similarity to generate diverse samples, this module can boost the model's ability to adapt to data distribution shifts.

5.4.4. *Effectiveness of diffusion reconstruction*

This section investigates the influence of diffusion reconstruction on the overall performance of the SMCD framework by removing this module (denoted as SMCD_4). As illustrated in Table 4, the experimental results demonstrate that SMCD is consistently superior to SMCD_4 across all datasets. On the ACM dataset, compared to SMCD_4, SMCD achieves an average Macro-F1 improvement of 7.51% and an average Micro-F1 improvement of 7.43%; on the DBLP dataset, the average Macro-F1 and Micro-F1 increase by 5.57% and 2.34%, respectively; on the IMDB dataset, the average Macro-F1 and Micro-F1 of SMCD are 3.8% and 4.6% higher than those of SMCD_4. These results indicate that diffusion reconstruction can enhance the quality and stability of embedding representations by recovering valid information from noise.

5.5. *Hyper-parameter Analysis*

This section explores the sensitivity of several key hyperparameters of SMCD through hyperparameter analysis. Specifically, we analyze the impact of value changes in the hyperparameters that control the contribution of auxiliary meta-paths (λ), the meta-path masking ratio (ϑ), the invariance-covariance contribution (δ), the hyperparameter governing covariance contribution (θ_1), the hyperparameter for controlling the distance between labeled and unlabeled classes (θ_2), the hyperparameter adjusting the subgraph size of meta-paths (k), and the number of diffusion steps (T) on node classification performance. To ensure statistical

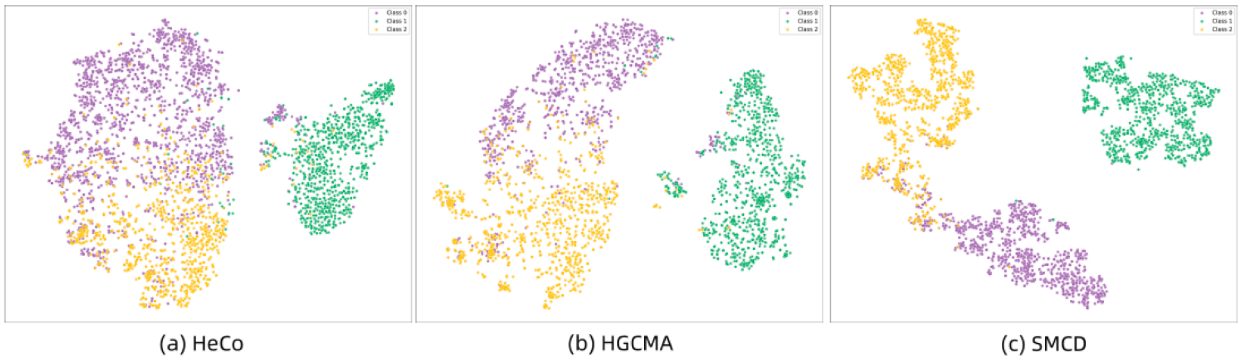


Fig. 6. Visualization of Different Models.

reliability, each parameter configuration is independently run 10 times, and the average Macro-F1 and Micro-F1 metrics are finally reported.

Parameter λ. The hyperparameter λ is used to control the contribution degree of auxiliary meta-paths. We set it to 0.1, 0.3, 0.5, 0.8, and 1.0 to assess the SMCD model's performance on the node classification task. As depicted in Fig. 2(a), the experimental results show that the model performance exhibits almost no fluctuation with the increase of λ. This demonstrates that the model has extremely strong robustness on datasets with regular data structures and high meta-path distinguishability (such as ACM and DBLP). In practical applications, there is no need to consume additional computing resources for hyperparameter tuning, and any value between 0.1 and 1.0 can be selected.

Parameter ϑ. The hyperparameter ϑ controls the meta-path masking ratio. We set it to 0.1, 0.3, 0.5, 0.7, and 0.9 to assess the SMCD model's performance on the node classification task. As shown in Fig. 2(b), the experimental results demonstrate the following: model performance on the ACM and DBLP datasets exhibits a slow downward trend as ϑ increases; in contrast, performance on the IMDB dataset fluctuates significantly and achieves relatively better results when ϑ is around 0.5. This indicates that for datasets highly dependent on meta-path information (e.g., DBLP), a lower meta-path masking ratio is preferred to ensure performance; for datasets with complex structures (e.g., IMDB), targeted tuning of ϑ is required to balance the impact of masking on feature capture.

Parameter δ. The hyperparameter δ is used to adjust the weight ratio between the invariance-covariance loss and the class-guided loss. We set its values to $\{0.1, 0.3, 0.5, 0.8, 1.0\}$ to assess the SMCD model's performance on the node classification task. As depicted in Fig. 2(c), the experimental findings demonstrate that: The model performance generally shows an upward trend with the increase of δ, which demonstrates that increasing the weight ratio of the invariance-covariance loss can effectively enhance the node classification capability of the SMCD model. The model achieves better performance when δ is in the range $[0.8, 1.0]$, indicating that the optimal weight ratio of the invariance-covariance loss lies within this interval.

Parameter θ_1 ***and*** θ_2. The parameter θ_1 and θ_2 are employed to control the covariance contribution and the distance between labeled and unlabeled classes, respectively. These two hyperparameters are varied within the range of $[0.1, 0.9]$ to evaluate the performance of the SMCD model on the node classification task. As illustrated in Fig. 3, the experimental results demonstrate that the SMCD model exhibits low sensitivity to these two hyperparameters (θ_1 and θ_2). When their values are adjusted within the interval of 0.1 to 0.9, the model performance only undergoes minor fluctuations. Therefore, there is no need to incur substantial costs for elaborate fine-tuning of these two parameters in practical applications.

Parameter k *and* T. The parameter k and T are used to adjust the meta-path subgraph size and the number of diffusion steps, respectively. To evaluate the performance of the SMCD model on the node classification task, we set k to 1, 2, 3, 4, 5 and T to 10, 50, 80, 100, 150. As shown in Fig. 4, the experimental results demonstrate that: when adjusted within the above value ranges, the Macro-F1 and Micro-F1 scores of the ACM and DBLP datasets remain consistently high (above 95), while the F1 metrics of the IMDB dataset only fluctuate slightly-no significant change is observed in the overall performance. However, the computational cost of the SMCD model increases as the values of k and T grow. Considering the balance between computational cost and performance, we set $k = 2$ and $T = 50$ in our final configuration.

5.6. Robustness Analysis

To investigate the model's robustness against data noise, we assess the magnitude of performance degradation by setting different noise ratios. Specifically, we randomly replace 10%, 30%, and 50% of the original samples across various relationship types with noise signals. We then evaluate three models-SMCD, HGCMA, and HeCo-on the ACM dataset and report the Macro-F1 and Micro-F1 metrics. As presented in Table 5, the experimental results demonstrate that the SMCD model exhibits a notably smaller performance degradation compared to the baseline models. This outcome confirms that the SMCD method possesses stronger denoising capability and can stably maintain favorable model performance in noisy data environments.

5.7. Scalability Analysis

To compare the runtime efficiency differences between SMCD and other baseline models following the multi-graph paradigm, we reproduce three baseline models-DMGI, HORACE [56], and HGCML-using an implementation framework identical to that of SMCD. Experiments are conducted under a unified hardware environment to ensure fair comparison. Fig. 5 shows the average complete running time of each model after 100 training rounds on the ACM dataset when the ratio of the number of nodes to the number of original samples changes. The results show that SMCD consistently achieves the shortest runtime, and as the ratio increases, the growth rate of its runtime is notably lower than that of other baseline models, demonstrating superior computational efficiency and scalability.

5.8. Visualization

To further evaluate the quality of learned embeddings, we perform a visualization analysis on the embeddings learned from the ACM dataset. Specifically, we employ the t-SNE[57] algorithm to plot the embedding distributions of the paper nodes of three models-HeCo, HGCMA, and SMCD-as illustrated in Fig. 6. It is evident that although the embedding clusters of all baseline models exhibit relatively clear boundaries,

SMCD achieves the most well-separated clustering effect compared to the baseline methods. Notably, SMCD further distinguishes the purple cluster from the yellow cluster, resulting in lower inter-cluster overlap. This outcome also explains why the proposed method attains superior performance.

6. Conclusion

This paper proposes a heterogeneous graph learning method (SMCD) that integrates semantic-aware meta-path perturbation and self-supervised and supervised collaborative optimization. First, auxiliary meta-paths are built based on the original meta-paths, and a dual augmentation scheme (semantic-level and task-level) is designed to generate diverse views. The semantic-level augmentation performs edge perturbation based on semantic similarity and enhances the semantics of core meta-paths with the help of a diffusion model; the task-level augmentation selects top-k relevant nodes using the diffusion model and semantic weights, thereby reconstructing local neighborhoods and the original graph structure. Second, a two-stage attention aggregation graph encoder is adopted to fully fuse structural and semantic information and output high-quality node embeddings. Finally, a self-supervised and supervised collaborative optimization strategy that adapts to label distribution is proposed, which balances the discriminability and generality of representations and adapts to scenarios with different degrees of label scarcity. By innovatively combining semantic enhancement and collaborative optimization, this paper provides a new paradigm for semantic mining and quality improvement of representation in heterogeneous graph learning. The effectiveness of SMCD has been fully validated through extensive experiments conducted on three public datasets. Future research can further extend to dynamic heterogeneous graph scenarios and explore the adaptive generation mechanism of meta-paths.

CRediT authorship contribution statement

Guanghua Ding: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization; **Rui Tang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization; **Xian Mo:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Y. Dong, N.V. Chawla, A. Swami, metapath2vec: Scalable representation learning for heterogeneous networks, in: Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining, 2017, pp. 135–144.
[2] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, P.S. Yu, Heterogeneous graph attention network, in: The World Wide Web Conference, 2019, pp. 2022–2032.
[3] P. Yu, C. Fu, Y. Yu, C. Huang, Z. Zhao, J. Dong, Multiplex heterogeneous graph convolutional network, in: Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining, 2022, pp. 2377–2387.
[4] X. Wang, N. Liu, H. Han, C. Shi, Self-supervised heterogeneous graph neural network with co-contrastive learning, in: Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining, 2021, pp. 1726–1736.
[5] S. Fan, J. Zhu, X. Han, C. Shi, L. Hu, B. Ma, Y. Li, Metapath-guided heterogeneous graph neural network for intent recommendation, in: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining, 2019, pp. 2478–2486.
[6] X. Mo, J. Pang, Z. Zhao, Attention-aware graph contrastive learning with topological relationship for recommendation, Applied Soft Computing 174 (2025) 113008.
[7] B. Hu, C. Shi, W.X. Zhao, P.S. Yu, Leveraging meta-path based context for top-n recommendation with a neural co-attention model, in: Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining, 2018, pp. 1531–1540.
[8] Q. Lv, M. Ding, Q. Liu, Y. Chen, W. Feng, S. He, C. Zhou, J. Jiang, Y. Dong, J. Tang, Are we really making much progress? revisiting, benchmarking and refining heterogeneous graph neural networks, in: Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining, 2021, pp. 1150–1160.
[9] X. Fu, J. Zhang, Z. Meng, I. King, Magnn: Metapath aggregated graph neural network for heterogeneous graph embedding, in: Proceedings of the Web Conference 2020, 2020, pp. 2331–2341.
[10] G. Zhu, Z. Zhu, H. Chen, C. Yuan, Y. Huang, Hagnn: Hybrid aggregation for heterogeneous graph neural networks, IEEE Transactions on Neural Networks and Learning Systems (2024).
[11] C. Zhang, D. Song, C. Huang, A. Swami, N.V. Chawla, Heterogeneous graph neural network, in: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining, 2019, pp. 793–803.
[12] S. Sang, K.-J. Chen, et al., HeGMN: Heterogeneous Graph Matching Network for Learning Graph Similarity, (2025). arXiv:2503.08739
[13] X. Yang, M. Yan, S. Pan, X. Ye, D. Fan, Simple and efficient heterogeneous graph neural network, in: Proceedings of the AAAI conference on artificial intelligence, 37, 2023, pp. 10816–10824.
[14] C. Zhang, D. Song, C. Huang, A. Swami, N.V. Chawla, Heterogeneous graph neural network, in: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining, 2019, pp. 793–803.
[15] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, Y. Bengio, Graph attention networks, (2017). arXiv:1710.10903
[16] K. Sun, Z. Lin, Z. Zhu, Multi-stage self-supervised learning for graph convolutional networks on graphs with few labeled nodes, in: Proceedings of the AAAI conference on artificial intelligence, 34, 2020, pp. 5892–5899.
[17] X. Chen, Y. Wang, J. Fang, Z. Meng, S. Liang, Heterogeneous graph contrastive learning with metapath-based augmentations, IEEE Transactions on Emerging Topics in Computational Intelligence 8 (1) (2023) 1003–1014.
[18] Y. Sun, D. Zhu, Y. Wang, Y. Fu, Z. Tian, GTC: GNN-transformer co-contrastive learning for self-supervised heterogeneous graph representation, Neural Networks 181 (2025) 106645.
[19] M.K. Hayat, S. Xue, J. Wu, B. Khan, J. Yang, Self-supervised Time-aware Heterogeneous Hypergraph Learning for Dynamic Graph-level Classification, in: Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, 2025, pp. 213–221.
[20] W. Du, H. Sun, H. Gao, G. Li, Y. Li, Contrastive Auxiliary Learning with Structure Transformation for Heterogeneous Graphs, in: Proceedings of the AAAI Conference on Artificial Intelligence, 39, 2025, pp. 16399–16407.
[21] Z. Wang, Q. Li, D. Yu, X. Han, X.-Z. Gao, S. Shen, Heterogeneous graph contrastive multi-view learning, in: Proceedings of the 2023 SIAM international conference on data mining (SDM), SIAM, 2023, pp. 136–144.
[22] B. Jing, S. Feng, Y. Xiang, X. Chen, Y. Chen, H. Tong, X-GOAL: Multiplex heterogeneous graph prototypical contrastive learning, in: Proceedings of the 31st ACM international conference on information & knowledge management, 2022, pp. 894–904.
[23] Y. Sun, J. Han, X. Yan, P.S. Yu, T. Wu, Pathsim: Meta path-based top-k similarity search in heterogeneous information networks, Proceedings of the VLDB Endowment 4 (11) (2011) 992–1003.
[24] Y. Sun, D. Zhu, Y. Wang, Y. Fu, Z. Tian, GTC: GNN-transformer co-contrastive learning for self-supervised heterogeneous graph representation, Neural Networks 181 (2025) 106645.
[25] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, P.S. Yu, Heterogeneous graph attention network, in: The world wide web conference, 2019, pp. 2022–2032.
[26] H. Hong, H. Guo, Y. Lin, X. Yang, Z. Li, J. Ye, An attention-based graph neural network for heterogeneous structural learning, in: Proceedings of the AAAI conference on artificial intelligence, 34, 2020, pp. 4132–4139.
[27] Z. Hu, Y. Dong, K. Wang, Y. Sun, Heterogeneous graph transformer, in: Proceedings of the web conference 2020, 2020, pp. 2704–2710.
[28] S. Yun, M. Jeong, R. Kim, J. Kang, H.J. Kim, Graph transformer networks, Advances in neural information processing systems 32 (2019).
[29] S. Zhu, C. Zhou, S. Pan, X. Zhu, B. Wang, Relation structure-aware heterogeneous graph neural network, in: 2019 IEEE international conference on data mining (ICDM), IEEE, 2019, pp. 1534–1539.
[30] P. Yu, C. Fu, Y. Yu, C. Huang, Z. Zhao, J. Dong, Multiplex heterogeneous graph convolutional network, in: Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining, 2022, pp. 2377–2387.
[31] C. Fu, G. Zheng, C. Huang, Y. Yu, J. Dong, Multiplex heterogeneous graph neural network with behavior pattern modeling, in: Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining, 2023, pp. 482–494.
[32] Y. Jiang, Z. Liu, Y. Song, Y. Shan, T. Wang, D. Yang, DAG-HFC: Dual-domain attention and graph optimization network for heterogeneous graph feature completion, Expert Systems with Applications 278 (2025) 127293.
[33] Y. Wang, Y. Liu, N. Liu, R. Miao, Y. Wang, X. Wang, AdaGCL+: An Adaptive Subgraph Contrastive Learning Towards Tackling Topological Bias, IEEE Transactions on Pattern Analysis and Machine Intelligence (2025).

[34] X. Shen, Y. Wang, K. Zhou, S. Pan, X. Wang, Optimizing ood detection in molecular graphs: A novel approach with diffusion models, in: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024, pp. 2640–2650.
[35] P. Veli, W. Fedus, W.L. Hamilton, P. Liò, Y. Bengio, D. Hjelm, et al., Deep graph infomax, in: International Conference on Learning Representations, 2019.
[36] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, L. Wang, Deep graph contrastive representation learning, (2020). arXiv:2006.04131
[37] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, L. Wang, Graph contrastive learning with adaptive augmentation, in: Proceedings of the web conference 2021, 2021, pp. 2069–2080.
[38] Y. Zhu, Y. Xu, H. Cui, C. Yang, Q. Liu, S. Wu, Structure-enhanced heterogeneous graph contrastive learning, in: Proceedings of the 2022 SIAM International Conference on Data Mining (SDM), SIAM, 2022, pp. 82–90.
[39] X. Jiang, Y. Lu, Y. Fang, C. Shi, Contrastive pre-training of GNNs on heterogeneous graphs, in: Proceedings of the 30th ACM international conference on information & knowledge management, 2021, pp. 803–812.
[40] X. Wang, N. Liu, H. Han, C. Shi, Self-supervised heterogeneous graph neural network with co-contrastive learning, in: Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining, 2021, pp. 1726–1736.
[41] Z. Wang, Q. Li, D. Yu, X. Han, X.-Z. Gao, S. Shen, Heterogeneous graph contrastive multi-view learning, in: Proceedings of the 2023 SIAM international conference on data mining (SDM), SIAM, 2023, pp. 136–144.
[42] C. Park, D. Kim, J. Han, H. Yu, Unsupervised attributed multiplex network embedding, in: Proceedings of the AAAI conference on artificial intelligence, 34, 2020, pp. 5371–5378.
[43] C. Wang, S. Zhou, K. Yu, D. Chen, B. Li, Y. Feng, C. Chen, Collaborative knowledge distillation for heterogeneous information network embedding, in: Proceedings of the ACM web conference 2022, 2022, pp. 1631–1639.
[44] Z. Wang, D. Yu, S. Shen, S. Zhang, H. Liu, S. Yao, M. Guo, Select your own counterparts: Self-supervised graph contrastive learning with positive sampling, IEEE Transactions on Neural Networks and Learning Systems 36 (4) (2024) 6858–6872.
[45] Z. Wang, Z. Zhang, N. Chawla, C. Zhang, Y. Ye, Gft: Graph foundation model with transferable tree vocabulary, Advances in Neural Information Processing Systems 37 (2024) 107403–107443.
[46] Z. Wang, Z. Zhang, T. Ma, N.V. Chawla, C. Zhang, Y. Ye, Towards Graph Foundation Models: Learning Generalities Across Graphs via Task-Trees, (2024). arXiv:2412.16441
[47] Z. Wang, Z. Zhang, T. Ma, C. Zhang, Y. Ye, Generative Graph Pattern Machine, in: The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2025.
[48] Y. Sun, J. Han, X. Yan, P.S. Yu, T. Wu, Pathsim: Meta path-based top-k similarity search in heterogeneous information networks, Proceedings of the VLDB Endowment 4 (11) (2011) 992–1003.
[49] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, Y. Bengio, et al., Graph attention networks, stat 1050 (20) (2017) 10–48550.
[50] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, P.S. Yu, Heterogeneous graph attention network, in: The World Wide Web Conference, 2019, pp. 2022–2032.
[51] T.N. Kipf, Semi-supervised classification with graph convolutional networks, (2016). arXiv:1609.02907
[52] J. Hu, B. Hooi, B. He, Efficient heterogeneous graph learning via random projection, IEEE Transactions on Knowledge and Data Engineering (2024).
[53] Z. Li, L. Xia, H. Hua, S. Zhang, S. Wang, C. Huang, DiffGraph: Heterogeneous Graph Diffusion Model, in: Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining, 2025, pp. 40–49.
[54] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, Y. Shen, Graph contrastive learning with augmentations, Advances in neural information processing systems 33 (2020) 5812–5823.
[55] S. Lloyd, Least squares quantization in PCM, IEEE transactions on information theory 28 (2) (1982) 129–137.
[56] Y. Zhu, Y. Xu, H. Cui, C. Yang, Q. Liu, S. Wu, Structure-enhanced heterogeneous graph contrastive learning, in: Proceedings of the 2022 SIAM International Conference on Data Mining (SDM), SIAM, 2022, pp. 82–90.
[57] M.L. van der, G. Hinton, Visualizing data using t-SNE, Journal of machine learning research 9 (Nov) (2008) 2579–2605.